



Semi-synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration by Inducing BAT Differentiation of FAPs

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INTRODUCTION

Fibro-adipogenic progenitor cells (FAPs) have been shown to be key regulators in repair of injured skeletal muscles. In uninjured muscles, FAPs are typically present in a quiescent state; however, they rapidly expand in response to muscle injury and can adopt a brown-adipose tissue (BAT) phenotype to support satellite cell myogenesis [1].

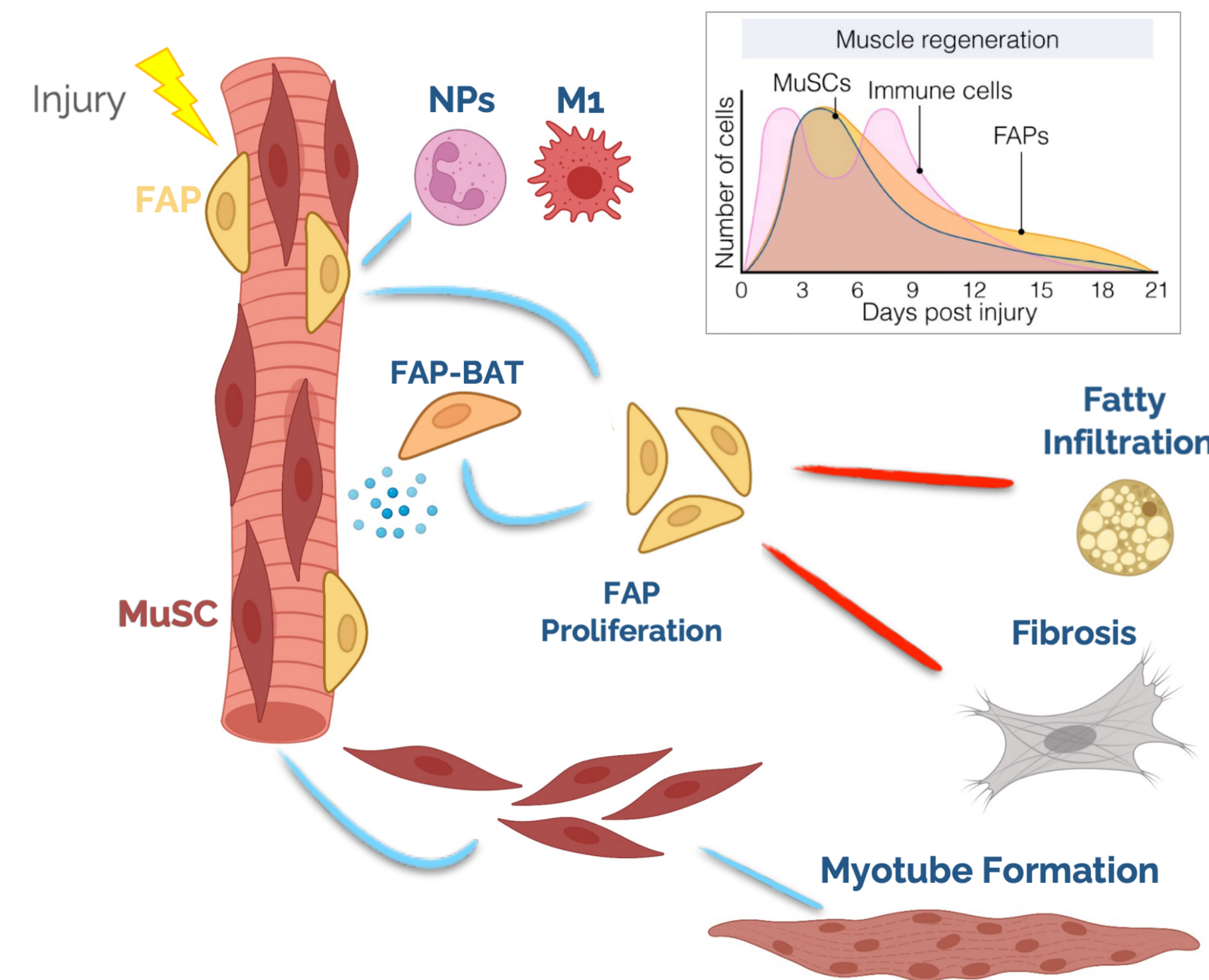


Figure 1. Schematic of cellular response after muscle injury. Inset: time course of cellular recruitment after muscle injury. Adapted from [2].

APPROACH

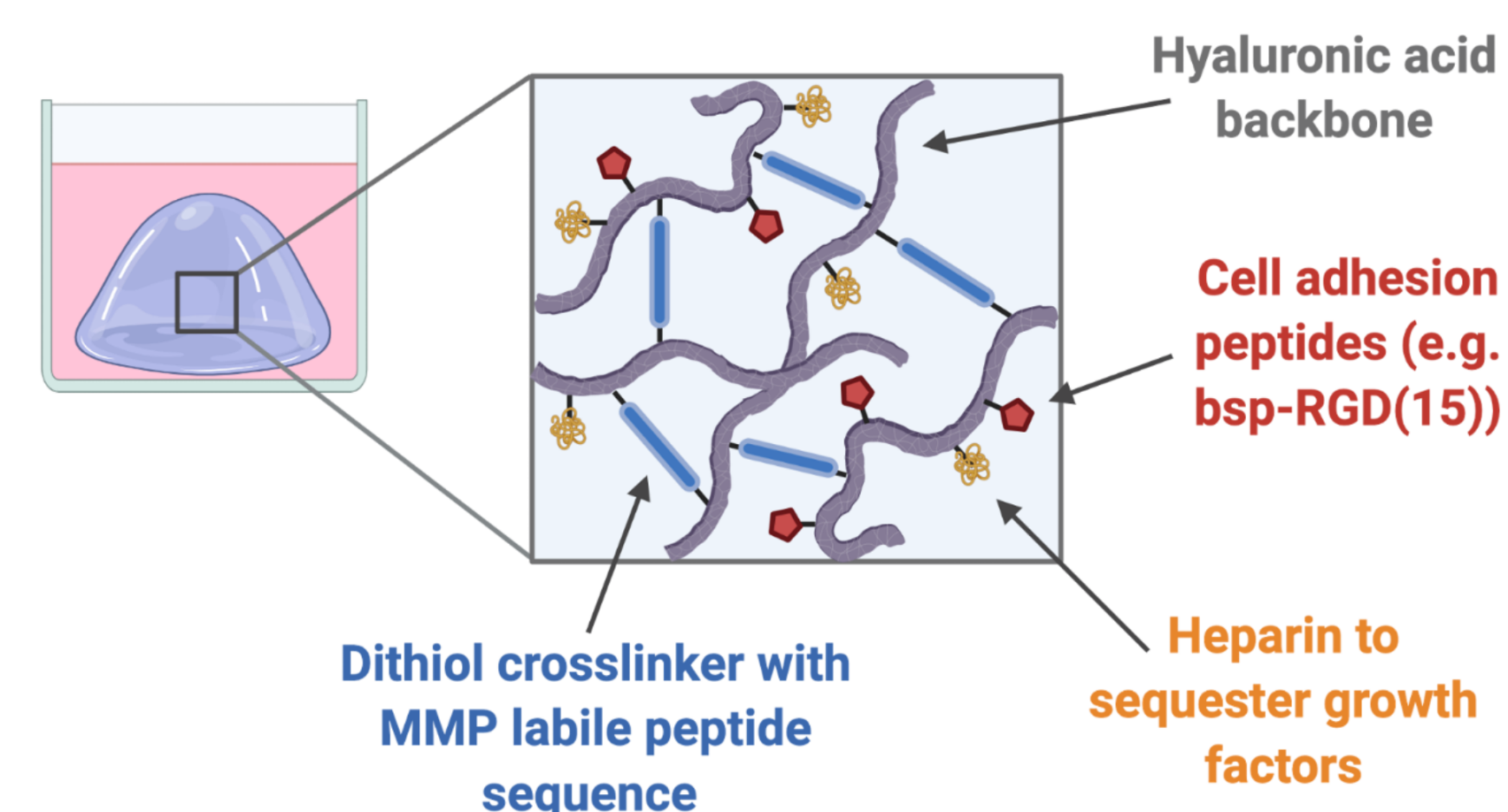


Figure 2. Semi-synthetic hyaluronic acid (HyA) hydrogel for tissue regeneration and stem cell transplantation.

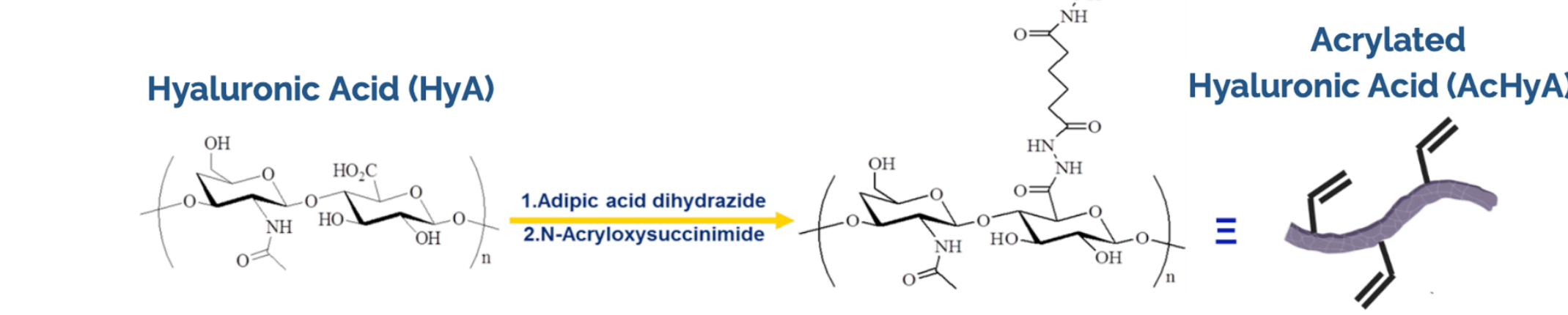
Highly tunable:

- Mechanical properties
- Degradation rate
- Cell specificity
- Growth factor sequestration

We hypothesize that our novel hydrogel would promote beige adipogenesis of human FAPs and thereby induce muscle regeneration.

METHODS

Synthesis of AchYA



Synthesis of HyA bioconjugates

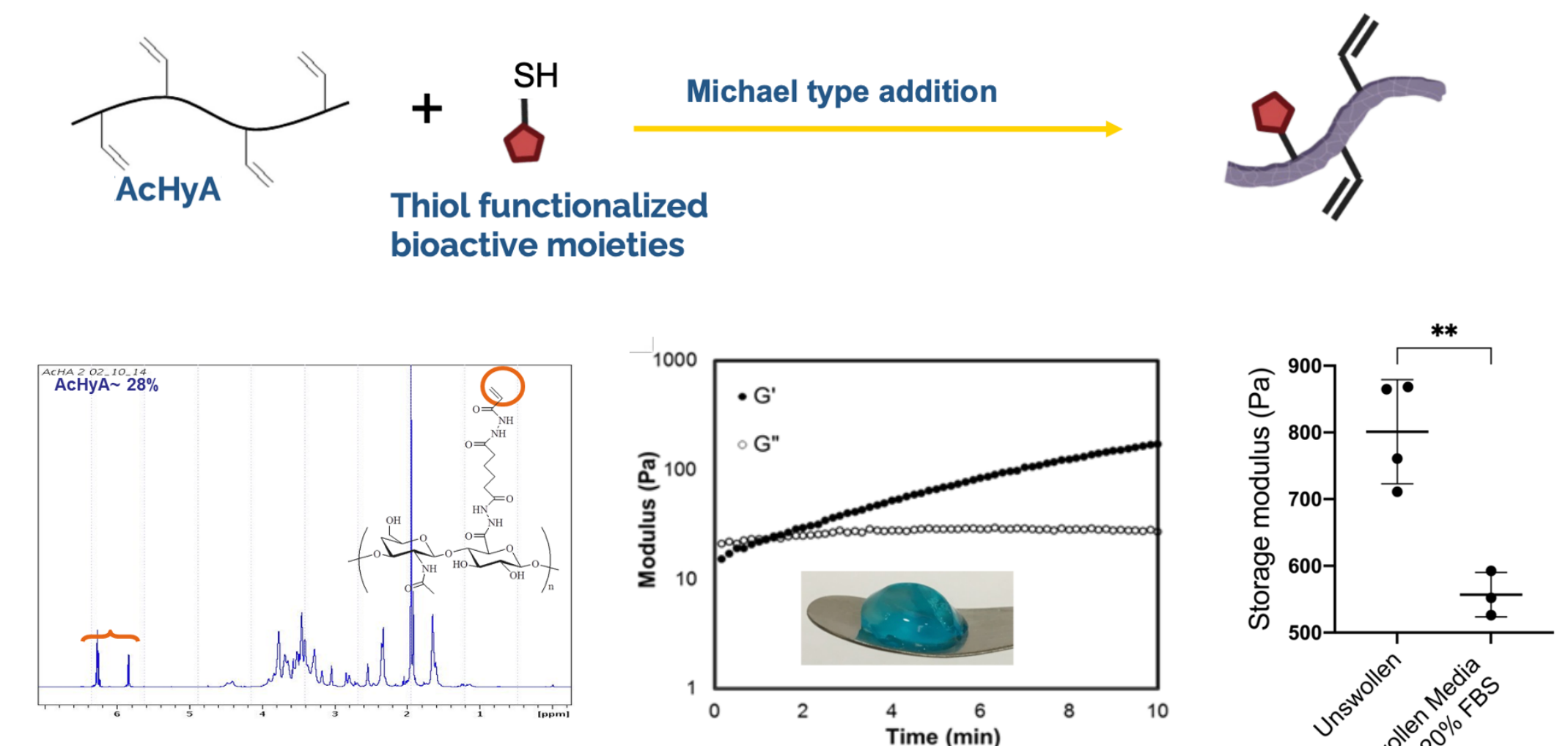


Figure 3. HyA macromer acrylation and synthesis of bioconjugates. The HyA hydrogels were synthesized using previously reported methods [3,4]. Briefly, HyA derivative carrying hydrazide groups (HyA-ADH) was synthesized and N-acryloxysuccinimide was subsequently reacted with the HyA-ADH solution to generate acrylate groups on the HyA (AchYA). For growth factor sequestration, SH-heparin was also synthesized to bind AchYA via Michael-type addition.

Deltoid muscle biopsies from RC tear patients

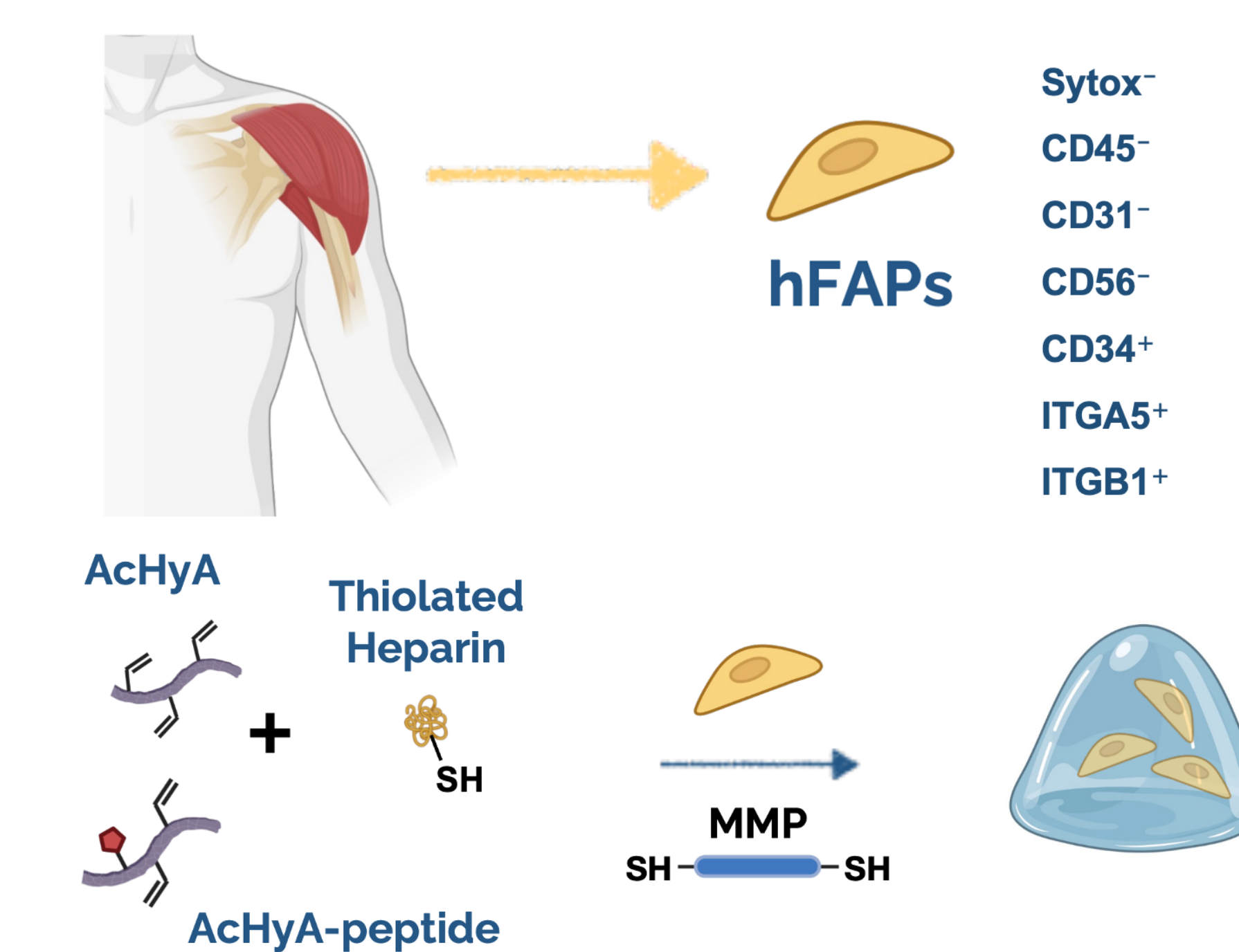


Figure 4. hFAP harvest and encapsulation. hFAPs were harvested from deltoid muscle biopsies (10 mg samples) performed on patients undergoing arthroscopic rotator cuff repair. Cells were encapsulated by mixing the cells into the macromer precursor solution along with one of four cell binding peptides (RGD, C16, Ag73, and P3). Gelation was initiated by addition of a bis-cysteine terminated peptide-cleavable crosslinker.

RESULTS

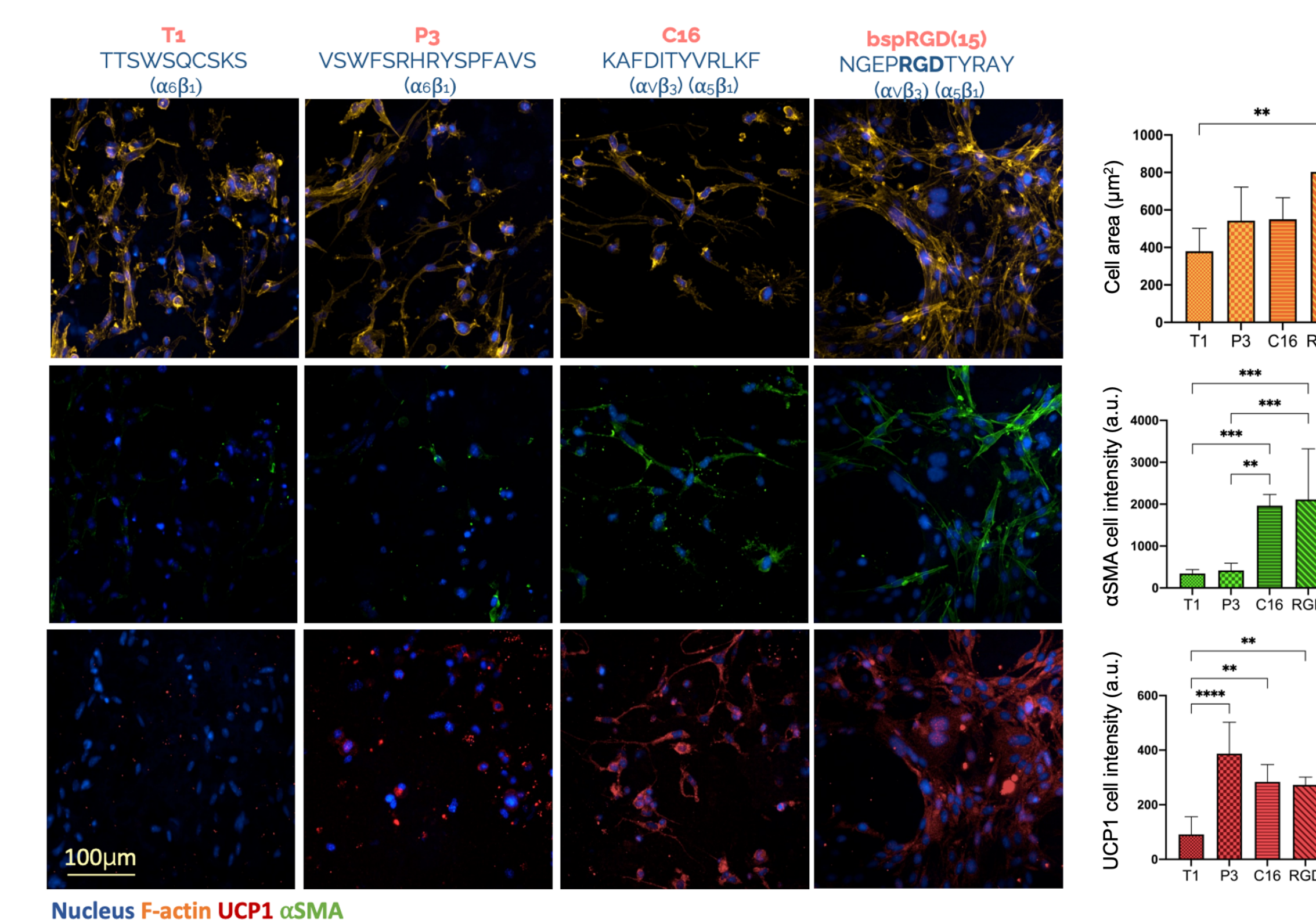


Figure 5. Culture of hFAPs in the HyA hydrogels with various integrin engaging peptides for 7 days. There was increased expression of the BAT marker UCP1 in the HyA-bsp-RGD(15), C16, and P3 peptide hydrogels compared to HyA-T1. αSMA, a fibrotic marker, was elevated in the HyA-bsp-RGD(15) and -C16 peptide hydrogels.

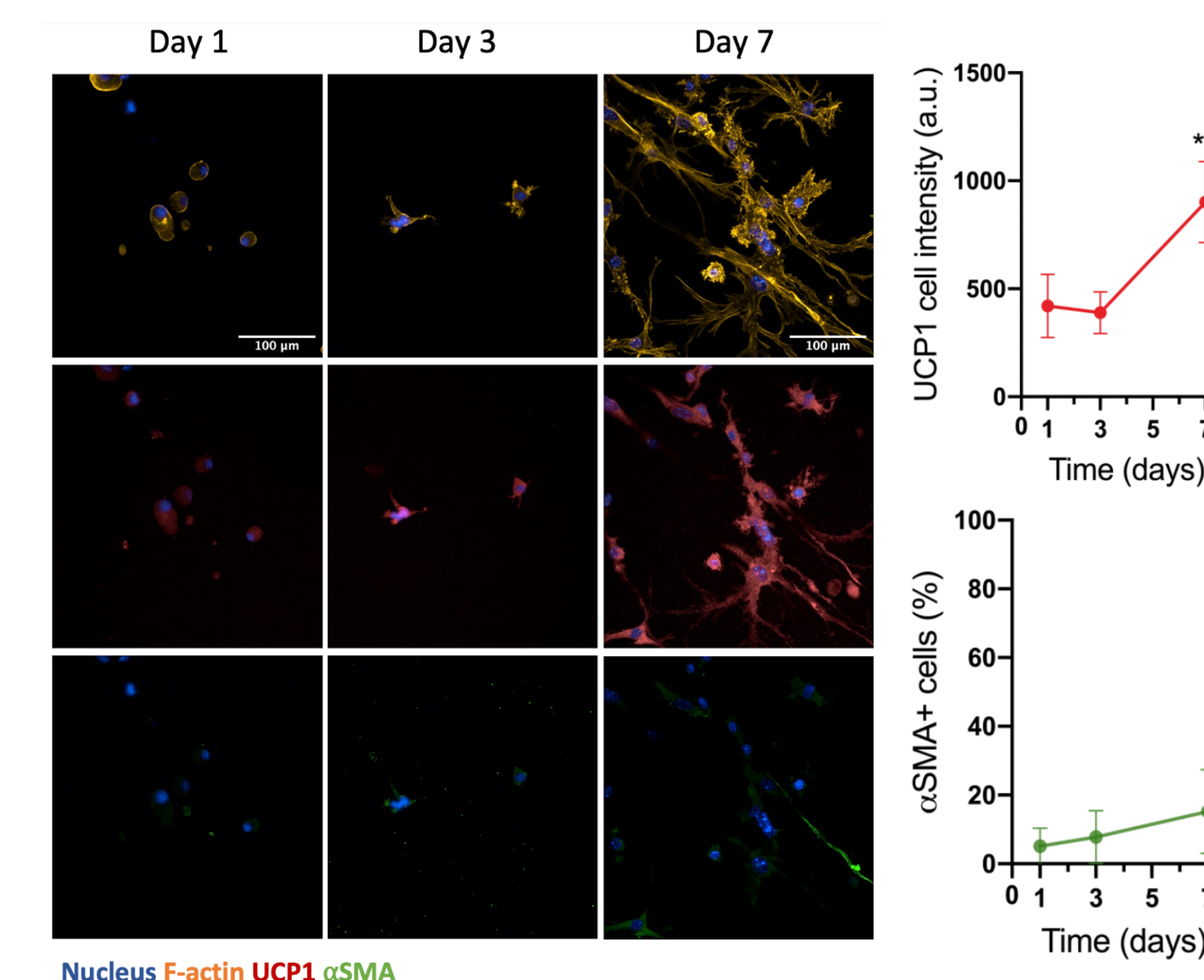


Figure 6. Time course of BAT phenotype. Immunostaining of hFAPs for UCP1 and αSMA after 1, 3 and 7 days in the HyA hydrogel with bsp-RGD(15). There was increased expression of the BAT marker UCP1, compared to day 1 and day 3. In contrast, αSMA, a fibrotic marker, did not increase significantly over time.

FUTURE DIRECTIONS

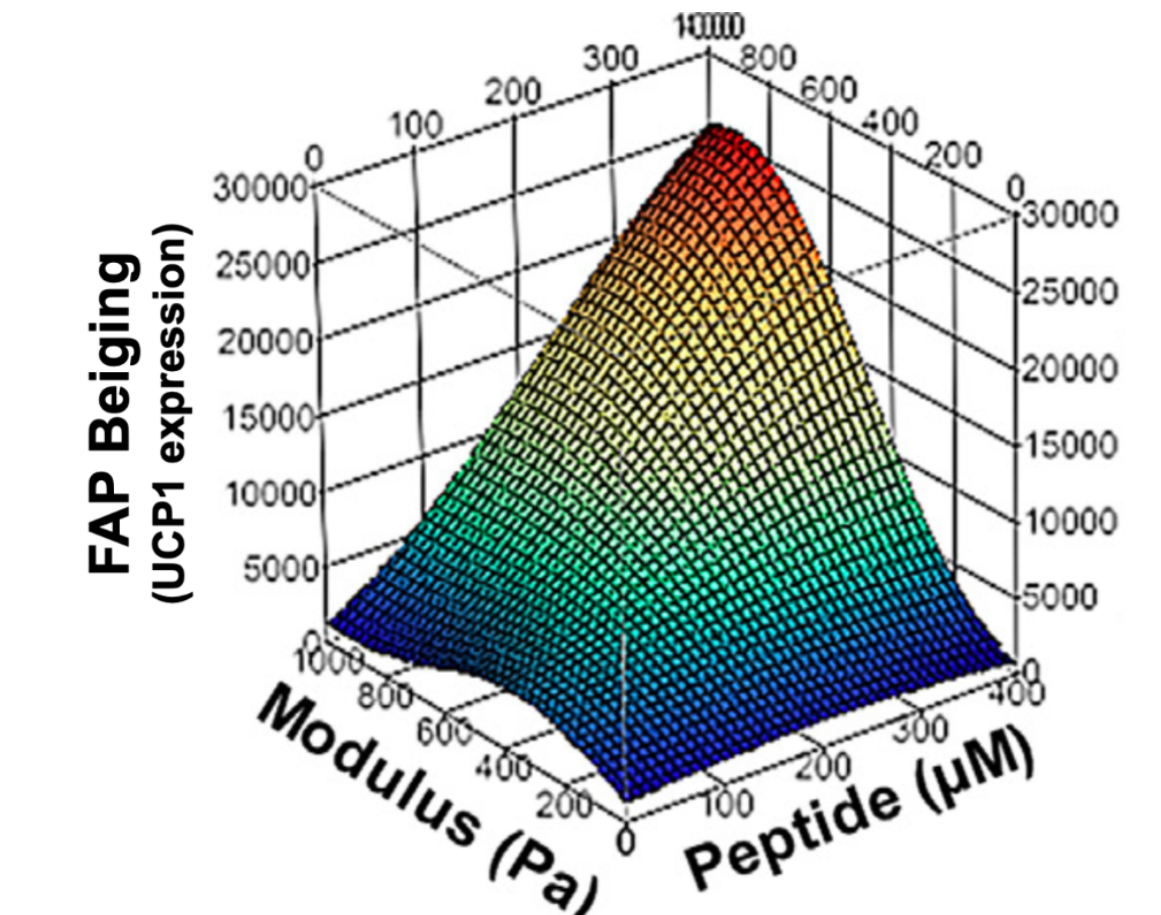


Figure 7. Representative response surface methodology (RSM) plot to optimize hydrogel formulation to maximize FAP-BAT differentiation as measured by UCP1 expression. Peptide variations can include both those for integrin engagement as well as the MMP-degradable crosslinker.

1. Continue hydrogel formulation optimization for optimal FAP-BAT differentiation and minimal fibrosis (Figure 7).
2. Conduct additional characterization of encapsulated hFAPs to verify BAT phenotype, characterize promyogenic activity (flow cytometry, cytokine secretion, RT-PCR, etc).
3. Perform co-encapsulations of hFAPs with muscle stem cells to evaluate hFAP-BAT-induced myotube fusion.
4. Treat hFAPs with M1/M2 macrophage media; macrophage co-encapsulations

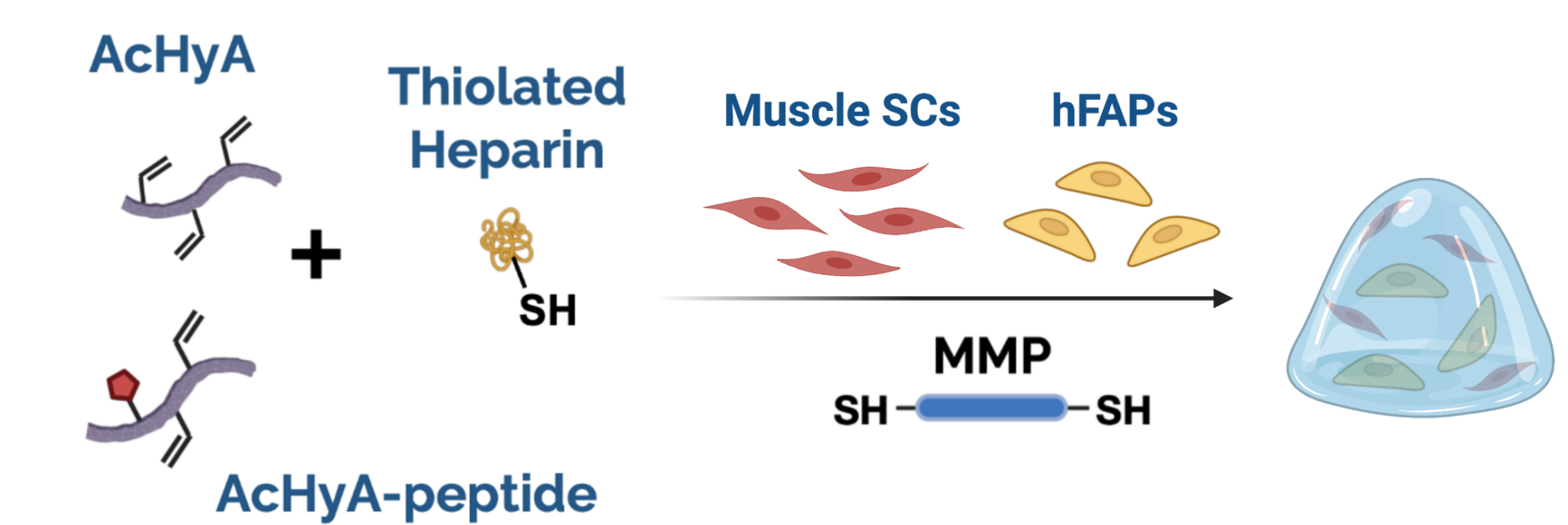


Figure 8. Schematic of co-encapsulation of hFAPs and muscle SCs into HyA hydrogels for myotube fusion analysis.

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REFERENCES

- [1] Liu X. et al., Muscles Ligaments Tendons J. 2016, 6(1), 6-15
- [2] M. Theret, Frontiers in Physiology, 12:673404, 2021
- [3] Jha et al., Biomaterials. 2015, 47, 1-12
- [4] Browne et al., ACS Biomater. Sci. Eng. 2020, 6(2), 1135-1143